MaskShift Supplementary Material

Anonymous

1 Scope

MaskShift is a benchmark/theory submission. It does not claim a new forecasting backbone, state-of-the-art clean forecasting accuracy, or a full official benchmark reproduction. The supplement records protocol details, dataset cards, mechanism cards, reviewer-response evidence, and reproduction commands for the MaskShift encoder-mask protocol.

2 Mechanism Generator Cards

- **mcar**. Independent random deletion at the target missing rate. Deployment read: Neutral packet loss or randomly sampled telemetry dropout. Scope guard: Control mechanism only; not an operational cause model.
- **block**. Contiguous channel-level missing spans with matched final rate. Deployment read: Short sensor logging gaps or temporary channel outages. Scope guard: Controls contiguity but not value dependence.
- **value_high**. Missing probability increases at high normalized values, then rate matching is enforced. Deployment read: Storm, peak-load, congestion, or saturation-related missingness. Scope guard: Synthetic proxy for value-dependent MNAR; not a causal claim.
- **volatility**. Missing probability increases with large local changes, then rate matching is enforced. Deployment read: Failure during unstable regimes, transitions, or high-frequency variation. Scope guard: Measures sensitivity to change-linked masks, not event detection quality.
- **blackout**. Contiguous time blocks remove many channels simultaneously with matched final rate. Deployment read: Power/network outage, site-level telemetry loss, or shared infrastructure failure. Scope guard: Does not model outage recovery dynamics beyond the encoder window.
- **retirement**. Selected channels disappear late in the sequence with matched final rate. Deployment read: Permanent sensor retirement, meter replacement, or channel decommissioning. Scope guard: Reported separately so the paper is not driven only by an obvious failure case.

3 Dataset Cards

Dataset	Rows	Channels	Natural missing	Max rate error	Evidence role
Weather	7000	21	0.00%	0.21%	positive core evidence
Electricity	7000	24	0.00%	0.19%	positive core evidence
Traffic	7000	24	0.00%	0.19%	boundary/mixed evidence
AirConvection	7000	8	20.08%	0.39%	boundary/mixed evidence

4 Core Configuration

The default MaskShift configuration is lookback 48, horizon 12, stride 8, target missing rate 35

5 Official PatchTST/TimeXer Coverage

Dataset	Source	Max degradation	Worst tau	ANOVA p	Gate
Weather	M9	123.7%	-1.000	0.0000	PASS
Electricity	M9	87.8%	-1.000	0.0023	PASS
Traffic	M16	35.4%	1.000	0.0854	MIXED/NEGATIVE
AirConvection	M16	46.1%	1.000	0.0001	MIXED/NEGATIVE

Weather and Electricity provide the positive official-architecture rank-reversal evidence. Traffic and AirConvection are mixed/negative for rank reversal and are retained as boundary evidence.

6 Reviewer Concern Matrix

ID	Concern	Evidence	Status
1	Is this just another missing-value model paper?	PAPER Sections 1/3/6; M5	Resolved by scope discipline.
2	Were modern forecasting backbones tested?	M9, M10, M16 summaries	Resolved as official-architecture adaptation.
3	Were missing-aware architectures tested?	M11, M12, M14 summaries	Resolved with mixed/negative evidence disclosed.
4	Are Weather/Electricity cherry-picked?	M16 table	Resolved by visible boundary evidence.
5	Is the result only sensor retirement?	M8 summary	Resolved for the core positive datasets.
6	Are relative degradation ratios unstable?	M7/M10 corrected metrics	Resolved by corrected reporting.
7	Does the typed head work as a method?	M2/M3 summaries; PAPER 4.6	Resolved by de-scoping.
8	Is the statistical evidence only aggregate means?	M3, M10, M13 summaries	Resolved for submission, with limitations retained.
9	Is the official-code claim over-stated?	PAPER limitations; checklist	Resolved by wording guardrails.
10	Can a reviewer reproduce the package?	README; supplement	Resolved as local reproduction package.

7 Reproducibility Commands

```
python3 -m experiments.MaskShift.m0__mask__suite
python3 -m experiments.MaskShift.m1__mechanism__audit
python3 -m experiments.MaskShift.m2__typed__head
python3 -m experiments.MaskShift.m3__statistical__tests
python3 -m experiments.MaskShift.m6__deep__backbone__sweep
python3 -m experiments.MaskShift.m7__severity__curves
python3 -m experiments.MaskShift.m8__mechanism__decomposition
python3 -m experiments.MaskShift.m9__official__tslib__reproduction
python3 -m experiments.MaskShift.m10__submission__hardening
python3 -m experiments.MaskShift.m11__official__ctf__missing__baseline
python3 -m experiments.MaskShift.m12__official__s4m__baseline
python3 -m experiments.MaskShift.m13__hierarchical__bootstrap
python3 -m experiments.MaskShift.m14__s4m__scale__validation
python3 -m experiments.MaskShift.m15__final__integrity__audit
python3 -m experiments.MaskShift.m16__official__tslib__full__coverage
python3 -m experiments.MaskShift.m17__submission__supplement
python3 -m experiments.MaskShift.m5__main__track__audit
```

8 External Revisions

The official-architecture adaptations are audited against TSLib revision `4e938a1`, ChannelTokenFormer revision `b1c100e`, and S4M revision `a718823`. These are architecture adaptations under MaskShift windows and masks, not full reproductions of the original papers' benchmark protocols.

9 Integrity Status

M15 reports `PASS_FINAL_INTEGRITY`. M5 reports `STRONG_CONFERENCE_READY` after requiring M15, M16, and this M17 supplement package.